1

Basic probability and statistics concepts

1.1. Probability

1.1.1. Random variables and their probability distributions

Suppose, in an experiment, events A, B or C are the possible outcomes. The **probability** of A is the chance of event A as an outcome, expressed as a number in the range [0,1], denoted by $\Pr(A)$. Similarly, $\Pr(B)$ is the probability of B and $\Pr(C)$ is the probability of C. Events A, B or C together form the complete **sample space** S of the experiment, such that $\Pr(S) = \Pr(A) + \Pr(B) + \Pr(C) = 1$. The **conditional probability** of A given B is given by (see e.g. Dekking et al. 2005, p. 26)

$$\Pr(A|B) = \frac{\Pr(A \wedge B)}{\Pr(C)} \tag{1.1}$$

where $\Pr(A \wedge B)$ is the probability of both A and B as an outcome. An event A is called **independent** of B if

$$\Pr(A|B) = \Pr(A) \tag{1.2}$$

If A is independent of B, than B is also independent of A and

$$\Pr(A \wedge B) = \Pr(A)\Pr(B) \tag{1.3}$$

In the above experiment, the possible events were expressed by A, B and C. A, B and C are letters. In computer algorithms, however, it is more convenient to express events of the sample space S as real numbers. Therefore, the possible outcomes are often described by a **random variable** X, which is a function that converts the events from the sample space S to real numbers in $\mathbb{R}$. If X can take on a (finite or infinite) countable number of values (e.g. either 1.0, 1.5, 2.0, or 2.5, with probabilities $\Pr(X = 1.0)$, $\Pr(X = 1.5)$, $\Pr(X = 2.0)$, and $\Pr(X = 2.5)$, respectively), X is called a **discrete random variable**. A random variable X is a **continuous random variable** if, for some function $p : \mathbb{R} \to \mathbb{R}$ and for any numbers a and b with $a \leq b$ (Dekking et al. 2005, Sect. 5.1)

$$\Pr(a \leq X \leq b) = \int_a^b p(x)dx, \tag{1.4}$$

where $p(x) \geq 0$ for all x and

$$\int_{-\infty}^{\infty} p(x)dx = 1. \tag{1.5}$$

Function p is called the **probability density function** of X. The probability density function describes the probability that X will take on a value between a and b. The **cumulative distribution function** P of a random variable X is the function $P : \mathbb{R} \to [0,1]$, defined by (Dekking et al. 2005, Sect. 4.2)

$$P(a) = \Pr(X \leq a) \quad \text{for } -\infty < a < \infty. \tag{1.6}$$

For a discrete random variable X which attains values $a_1, a_2, ...,$, the cumulative distribution function can be computed as

$$P(a) = \sum_{a_i \le a} \Pr(X = a_i) \tag{1.7}$$

For a continuous random variable X, the cumulative distribution function can be computed as

$$P(a) = \int_{-\infty}^{a} p(x)dx \tag{1.8}$$

The probability density function of a continuous random variable can be retrieved from its cumulative distribution function as follows (see Dekking et al. 2005, Sect. 5.1):

$$p(x) = \frac{d}{dx}P(x) \tag{1.9}$$

Both the probability density function and the cumulative distribution function of a continuous random variable X contain all the probabilistic information about X. That is, either of them fully describe the **probability distribution** of X. There are several commonly used probability distributions. For example, if we pick a random number between values c and d with equal probabilities for all numbers between c and d, a random variable X is said to be **uniformly distributed**. In that case, $p(x)$ of X is defined by (Dekking et al. 2005, Sect. 5.2):

$$p(x) = \begin{cases} \frac{1}{d-c} & \text{for } c \le x \le d, \\ 0 & \text{otherwise.} \end{cases} \tag{1.10}$$

In what follows, the uniform distribution with the probability density function of Eq. 1.10, is denoted by $U(c,d)$. Uniform distributions are often used in computer algorithms that model random processes, where **(pseudo) random number generators** are used to simulate $U(c,d)$ (see Sect. **??**). For example, if a random number has to be generated from a uniformly distributed random variable ranging between 0 and 1, it can readily be calculated from Eqs. 1.4 and 1.10 that the probability of picking a number between 0 and 0.5 equals 0.5:

$$\Pr(0 \le X \le 0.5) = \int_0^{0.5} p(x)dx = \int_0^{0.5} \frac{1}{1-0}dx = 0.5. \tag{1.11}$$

and, in general for $U(0,1)$ and some $q > 0$,

$$\Pr(0 \le X \le q) = q \tag{1.12}$$

Other common distributions are the **normal distribution** $N(\mu, \sigma^2)$, with probability density function (Dekking et al. 2005, Sect. 5.5):

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] \quad \text{for } -\infty < x < \infty, \tag{1.13}$$

and the **exponential distribution** $Exp(\lambda)$, with probability density function (Dekking et al. 2005, Sect. 5.3):

$$p(x) = \lambda \exp[-\lambda x] \quad \text{for } x \ge \infty. \tag{1.14}$$

Note that we do not only consider probability distributions of continuous random variables, but also of discrete random variables. For example, when a discrete random variable X can attain the value 1 (e.g. heads in a coin tossing experiment) or 0 (tails), X has a **Bernoulli distribution** $Ber(q)$ with parameter q, where $0 \le q \le 1$ and (Dekking et al. 2005, Sect. 4.3)

$$\Pr(X = 1) = q \quad \text{and} \quad \Pr(X = 0) = 1 - q \tag{1.15}$$

If the probability of throwing heads $\Pr(X = 1)$ equals the probability of throwing tails $\Pr(X = 0)$, then $q = 0.5$.

Sometimes, we are interested in the probability density of two random variables simultaneously. Continuous random variables X and Y have a **joint distribution** if for some function $p : \mathbb{R}^2 \to \mathbb{R}$ and for all number a_1, a_2 and b_1, b_2 with $a_1 \le b_1$ and $a_2 \le b_2$ (Dekking et al. 2005, p. 119)

$$P(a_1 \le X \le b_1, a_2 \le Y \le b_2) = \int_{a_1}^{b_1}\int_{a_2}^{b_2} p(x,y)dxdy \tag{1.16}$$

where $p(x, y) \geq 0$ for all x and y and

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x, y)\,dx\,dy = 1. \tag{1.17}$$

In Eq. 1.17, function p is called the **joint probability density function** of X and Y. The probability distributions of X only or Y only, also called **marginal probability density functions**, can readily be obtained by integrating out the other variable, that is (Dekking et al. 2005, p. 122):

$$p(x) = \int_{-\infty}^{\infty} p(x, y)\,dy \quad \text{and} \quad p(y) = \int_{-\infty}^{\infty} p(x, y)\,dx. \tag{1.18}$$

The **conditional probability density** of Y given X is defined as (see Rice 2006, Sect. 3.5.2)

$$p(y|x) = \frac{p(x, y)}{p(x)}, \tag{1.19}$$

if $0 < p(x) < \infty$, and 0 otherwise. Note that if $p(y|x) = p(y)$, continuous random variables X and Y are said to be **independent** (see Dekking et al. 2005, Eq. 9.6).

Bibliography

Dekking, F., Kraaikamp, C., Lopuhaa, H. and Meester, L. (2005), *A modern introduction to probability theory and statistics*, Vol. Springer texts in statistics, Springer.

Rice, J. A. (2006), *Mathematical Statistics and Data Analysis.*, third edn, Belmont, CA: Duxbury Press.